

T1K vs. Immuannot on the giraffe/whatshap 1000G pangenome-typing pipeline

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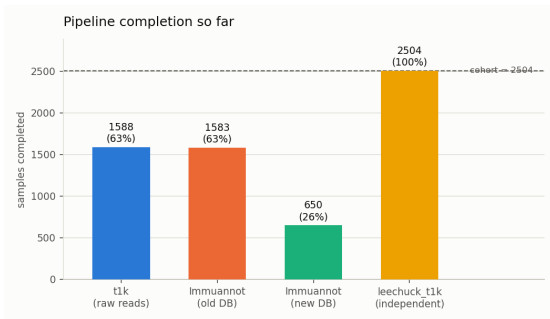
BioHackathon Japan 2026, HLA/pangenome group

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Four HLA typing runs on the same 1000G short-read cohort, compared pairwise

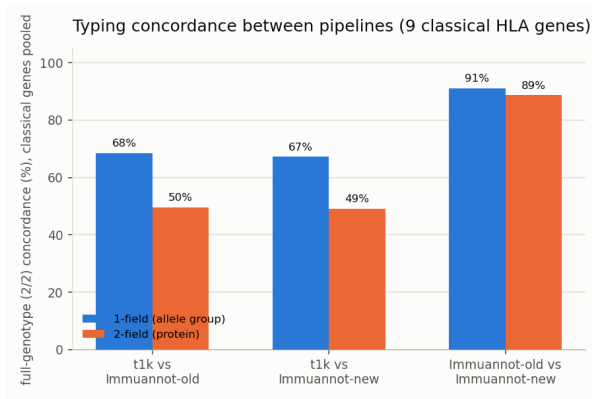
- ▶ **t1k**: run-t1k directly on raw FASTQs extracted per sample from the 1000G high-coverage CRAMs (`mhc_giraffe/t1k`); independent of the graph/phasing steps below
- ▶ **Immuannot, old DB**: `vg giraffe` alignment to leechuck's whole-MHC Minigraph-Cactus graph → `vg call` → WhatsHap read-backed phasing → `bcftools consensus` hap1/hap2 FASTAs → Immuannot against leechuck's frozen IPD-IMGT/HLA 3.55.0
- ▶ **Immuannot, new DB**: identical hap1/hap2 consensus FASTAs, re-annotated against a freshly built IPD-IMGT/HLA 3.65.0 (same release T1K uses) – isolates database-version effects only
- ▶ **leechuck_t1k**: leechuck's own, fully independent run-t1k on the same 2,504-sample cohort (`1000G_MHC/t1k`), same file format as t1k above, no pangenome/consensus step at all – a second opinion on t1k itself
- ▶ **Comparison**: unordered allele pairs, truncated to 1-field (allele group) and 2-field (protein) resolution; Immuannot's `:new` suffix (no exact database match) stripped before truncation, not treated as a literal field; “overall” figures pool only the 9 classical HLA genes (A/B/C/DRA/DRB1/DQA1/DQB1/DPA1/DPB1), not the full gene list (see slide 9)

t1k and Immuannot each cover roughly two-thirds of the cohort; leechuck_t1k is fully done



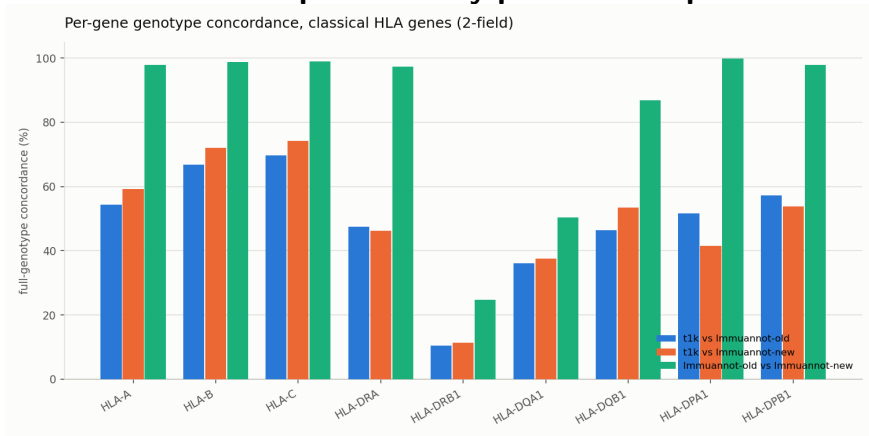
- ▶ t1k and Immuannot-old-DB run independently per sample and finish at the same pace ($\approx 63\%$); Immuannot-new-DB is a smaller re-run launched later, still catching up (26%); leechuck_t1k is a separate, already-completed 2,504-sample run
- ▶ All figures on later slides use whatever fraction of the cohort had finished at the time of analysis

Immuannot's old and new databases agree closely; t1k agrees with Immuannot about half the time



- ▶ Immuannot-old vs Immuannot-new: 89% full-genotype (2-field), 91% at 1-field, on the 9 classical genes – expected, since it is the same consensus sequence, only the reference database changed
- ▶ t1k vs Immuannot (either DB): $\approx$ 49–50% full-genotype concordance on the same 9 genes – Immuannot types off a phased short-read consensus (subject to phasing/assembly error), t1k types straight from reads. Confirmed to be a real gap, not a one-off t1k artifact (slide 8)

HLA-DRB1 concordance collapses in every pairwise comparison

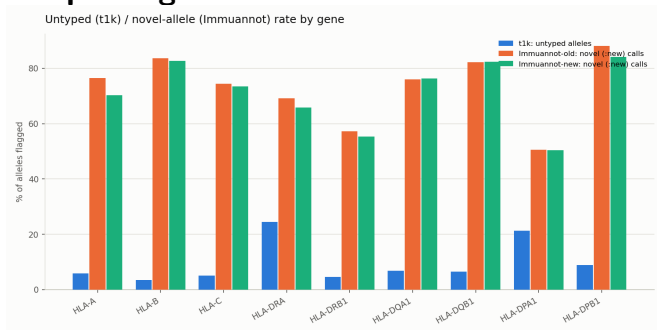


- ▶ A/B/C/DPA1/DPB1: Immuannot-old vs Immuannot-new $\geq 97\%$; t1k vs either Immuannot 42–74%
- ▶ HLA-DRB1: only 10–25% concordance for every pair, including Immuannot-old vs Immuannot-new (same consensus sequence!) – a paralog-confusable, hard-to-phase locus, not a database-version or typer artifact

How: genotype comparison at 1- and 2-field resolution

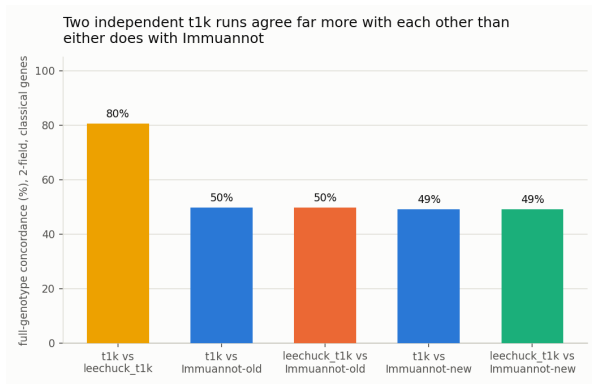
- ▶ Truncate each allele string to its first ($k=1$) or first two ($k=2$) colon-separated fields after the gene, e.g. HLA-A*29:02:01:01 $\rightarrow$ HLA-A*29:02
- ▶ Immuannot's :new suffix marks the point past which no exact database allele matched – HLA-A*01:new is only *1-field*-resolved, not a literal second field of “new”; stripped before truncation, and a 2-field comparison is skipped (not counted as a mismatch) when fewer than 2 real fields remain
- ▶ Genotype match: best of the two possible allele-pairings (hap1/hap2 vs allele1/allele2 order is arbitrary between pipelines); full match requires 2/2 alleles to agree
- ▶ Script: analysis/gc_typing_concordance.py; tables in results/tables/gc_*.tsv

70–88% of Immuannot's classical-gene calls are novel (:new), consistent with imperfect short-read phasing



- ▶ t1k rarely leaves a classical gene untyped (3–9%, DPA1/DRA higher at 21–24%); leechuck_t1k shows the same rates gene-for-gene, confirming this is a property of the reads, not a t1k-run artifact
- ▶ Immuannot flags a novel allele on 70–88% of A/B/C/DQB1/DPB1 calls, old and new DB alike – the consensus sequence itself, not the database, is the limiting factor (matches the pipeline's own documented caveat: WhatsHap phasing can leave chimeric/ambiguous consensus for highly polymorphic genes)

Two independent t1k runs agree far more with each other than either does with Immuannot

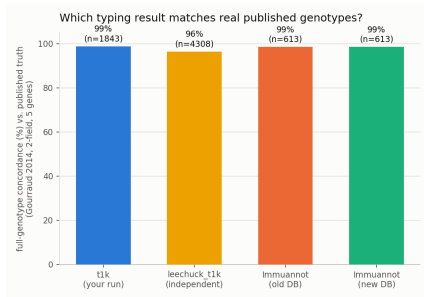


- ▶ leechuck_t1k vs this pipeline's t1k: 80% full-genotype concordance, 9 classical genes – driven down mainly by HLA-DRA/DPA1 (low-polymorphism), the other 7 genes are 82–93%
- ▶ leechuck_t1k vs Immuannot (either DB): 49–50% – essentially the same gap as this pipeline's own t1k shows vs Immuannot (previous slide)
- ▶ Conclusion: swapping in an independent t1k run barely moves the t1k-vs-Immuannot number – the discordance is about Immuannot/consensus quality, not which t1k run is used

Neither t1k run touches the pangenome graph – only Immuannot does

- ▶ `mhc_pipeline.sbatch` labels its own t1k step “independent of the graph/phasing steps above” (line 144) – both t1k runs (this pipeline’s and leechuck’s) type directly off raw FASTQs, no giraffe/vg/consensus step involved
- ▶ The *only* pangenome-derived arm in this whole comparison is Immuannot, which types the `vg giraffe → vg call → WhatsHap → bcftools consensus` sequence
- ▶ So “did the pangenome alignment improve typing” really means: does Immuannot-on-pangenome-consensus beat t1k? Checked directly against real published truth genotypes on the next slide (Gourraud et al. 2014 1000G HLA diversity panel, Sanger/SSO typing, 955 samples also in this cohort)

When Immuannot commits to a 2-field call, it's as accurate as t1k – but it commits far less often



- ▶ Tied on accuracy: t1k 98.8%, Immuannot 98.5% (either DB) full-genotype match to published truth (2-field, 5 classical genes)
- ▶ Denominators aren't comparable though: t1k resolves 90% of its truth-eligible calls to 2-field (n=1843); Immuannot only 33% (n=613) – the rest are :new/low-resolution calls (slide 7), uncheckable, not wrong
- ▶ So: pangenome alignment + phasing doesn't demonstrably change per-call accuracy – it costs *resolution/coverage*, not correctness

Correction: an earlier pass compared against a stale precomputed table

- ▶ First attempt used `leechuck/hla/typing/conc_partial/data/conc_t1k_vs_published.tsv` (412 samples, 5 genes) as “leechuck’s t1k” – it showed only 86% agreement with this pipeline’s t1k, and disagreements looked like a coverage/read-extraction effect (traced, incorrectly, to the 16 GRCh38 chr6 MHC ALT contigs missing from this pipeline’s read extraction)
- ▶ Checking that hypothesis directly on a discordant case (HG00097, HLA-B) found *zero* reads split onto an ALT contig at that locus, and identical read depth/content between the two extractions – the ALT-contig story does not hold up
- ▶ Root cause of the confusion: `conc_t1k_vs_published.tsv` predates leechuck’s current, live `1000G_MHC/t1k/` run (all 2,504 samples) and simply disagreed with leechuck’s *own current* output for the same sample/gene – it was never a fair comparison to begin with
- ▶ Re-run against the live `1000G_MHC/t1k/` directory instead (previous slide): agreement jumps to 80% on classical genes (94–99% on A/B/C/DRB1/DQB1 individually); the ALT-contig slides have been removed
- ▶ Lesson: always check a precomputed table’s timestamp/provenance against the pipeline that’s supposed to have produced it, especially when it disagrees with a live re-derivation

Take-home

- ▶ **Pangenome alignment did not improve typing accuracy here** – against real published truth, Immuannot-on-pangenome-consensus (98.5%) and t1k (98.8%) are statistically tied when Immuannot actually commits to an answer. The pangenome/phasing arm's real cost is *resolution*: it only reaches a checkable 2-field call 33% of the time vs. t1k's 90%
- ▶ Immuannot's database version (3.55 vs 3.65) barely matters (89–91% agreement, classical genes); the consensus sequence quality from graph alignment + WhatsHap phasing is the real bottleneck, especially for A/B/C/DQB1/DPB1 (70–88% flagged novel) and DRB1 (concordance collapses even DB-vs-DB)
- ▶ t1k-vs-Immuannot discordance ($\approx$ 49–50%, classical genes) is real and confirmed by a second, fully independent t1k run – not an artifact of one t1k invocation
- ▶ t1k remains the more practically useful call for benchmarking: same accuracy as Immuannot when both can be checked, but usable on far more calls
- ▶ Open: revisit DRB1 specifically (worst locus everywhere, including DB-vs-DB) once Immuannot-new-DB finishes the full cohort; understand why HLA-DRA/DPA1 show more t1k-run-to-run disagreement than A/B/C/DRB1/DQB1 do
- ▶ Script, figures, tables: `pangenome-bh26/hla/analysis/gc_*.py`, `results/figures/gc_*.png`, `results/tables/gc_*.tsv`